

Ashlynn Xavier

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EDUCATION

University of Rochester	Rochester, NY
<i>B.S. with Highest Distinction, Biomedical Engineering (Biosignals & Biosystems Concentration)</i>	<i>May 2025</i>
• Minor in Music Cognition • GPA 3.86 out of 4.00 • Campus Involvement and Leadership: Tau Beta Pi NY Kappa (Tutoring Chair), Off Broadway On Campus (Music Director, Fundraising Chair), UR Symphony Orchestra, UR Orchestras Committee (Co-Chair), Learn To Be (Volunteer Tutor)	

ENGINEERING AND RESEARCH EXPERIENCE

Feinstein Institutes for Medical Research	Manhasset, NY
<i>Junior Electrical/Systems Engineer, Bouton Lab (Part-Time)</i>	<i>Sep 2026 - Present</i>
• Continuing in an expanded role, with increased responsibility for experiment development, data analysis pipelines, and mentoring new lab personnel • Author and maintain SOPs and technical documentation to be used by clinicians and engineers • Correspond with external vendors via email and phone for equipment support to resolve hardware issues and minimize downtime in experiment data collection	
<i>Electrical/Systems Engineering Intern, Bouton Lab (Neural Bypass and Brain-Computer Interface)</i>	<i>Mar - Jun 2026</i>
• Developed, validated, and documented computer vision and signal processing pipelines to quantify changes in participant hand kinematics, collaborating with clinicians to identify meaningful metrics • Designed and iterated on 3D printed task assessment components in Fusion 360 based on collaboration with physical and occupational therapists, supporting stroke and spinal cord injury patient studies	
University of Rochester	Rochester, NY
<i>Research Intern, Norman-Haignere Lab (Computational Neuroscience of Audition)</i>	<i>Mar 2024 - May 2025</i>
• Designed and piloted a human participant experiment in compliance with IRB protocols, including programming graphical user-interfaces with JavaScript • Presented frequent updates to all stakeholders, updating procedures and documentation based on feedback	
<i>Teaching Assistant, Department of Biomedical Engineering</i>	<i>Feb 2024 - May 2025</i>
• Taught courses covering MATLAB coding, signals, analog circuits, statistics, and research ethics • Authored detailed rubrics and documentation to standardize grading between instructors and sections	
<i>Research Assistant, Maddox Lab (Auditory and Multisensory Processing)</i>	<i>October 2022 - December 2023</i>
• Supervised EEG experiments and ensured IRB and HIPAA compliance in data handling and informed consent • Independently set up and troubleshooted EEG equipment, batteries, connections to computer, and Python code	
Boston University Photonics Center	Boston, MA
<i>Research Intern, NSF REU in Integrated Nanomanufacturing</i>	<i>Jun - Aug 2023</i>
• Designed and validated an electrochemical biosensor prototype iteration using magnetic nanoparticles • Conducted a literature review and learned necessary procedures for project design within first two weeks • Characterized and validated design by analyzing and visualizing electrochemical analysis data in Python • Collaborated with stakeholders to improve design based on feedback, ensuring final product met team needs	

HONORS AND AWARDS

<i>Highest Distinction:</i> top 10% GPA in department	<i>Patrick P. Lee Foundation Scholarship:</i> student exemplifying integrity, leadership, and service
<i>Cum Laude:</i> top 15% GPA in graduating class	<i>Research and Innovation Grant:</i> strong potential in research and entrepreneurship
<i>Tau Beta Pi:</i> the Engineering Honor Society	
<i>Alpha Eta Mu Beta:</i> the Biomedical Engineering Honor Society	

PUBLICATIONS

M. J. Henry, ... R.K. Maddox, ... A. G. Xavier, ... J. E. Peele, "How strong is the rhythm of perception? A registered replication of Hickok et al. (2015)," Royal Society Open Science, vol. 12, no. 6, Jun. 2025.doi:10.1098/rsos.220497

SKILLS

- **Programming & Software:** Python (PyTorch, TensorFlow, NumPy, Pandas, SciPy, Ctypes), MATLAB, C++, HTML/CSS/JavaScript, GitHub, DLLs/SDKs
- **Prototyping & Device Design:** CAD/3D printing (SolidWorks, Fusion 360), soldering, circuit design, LTSpice, Arduino, bench testing/validation
- **Clinical & Human Participants Research:** IRB/HIPAA compliance, informed consent, experimental design, poster and slide presentations; manuscript, proposal, and SOP writing
- **Data Analysis:** digital and analog signal processing, statistics, data visualization, computer vision/audition
- **Electrophysiology:** EEG, brain-computer interfaces, knowledge of neurobiology, behavioral experiments
- **Communication:** SOP writing, stakeholder communication, MS Office/Google Suite, Adobe Illustrator

RELEVANT PROJECTS

Three-Dimensional Finger Tracking with RGB-Depth Cameras (NBBCI Lab)

- Developed an automated pipeline to quantify hand movement using RGB-depth cameras in Python at 20 FPS
- Built and validated pipeline design to achieve position measurements accuracy within 5 mm of error using an automated point cloud analysis script and ground truth objects

Electronic Wearable Non-Invasive Blood Glucose Monitor (University of Rochester)

- Prototyped a wearable electronic sensor and MATLAB algorithm for blood glucose level estimation based on needs obtained from stakeholder interviews
- Validated sensor and algorithm design based on data from 10 human participants across 30 readings
- Outlined a plan to meet relevant standards and FDA regulations
- Produced detailed design documentation to support future development teams

Voltage Regulator and Casing (NBBCI Lab)

- Soldered a voltage regulator circuit and tested connections with a multimeter to ensure safety/safe failing
- Designed a snap-fit enclosure in Fusion360 and 3D printed with a Creality printer

Enhanced Captioning for Musical Theater (University of Rochester)

- With a partner, created a captioning system in Python for audio with speech and song from multiple speakers
- Wrote a report following standard journal guidelines and presented our system to the class

Classifying EEG Signals with TensorFlow (Personal, In-Progress)

- Self-teaching machine learning concepts including model selection, building models in TensorFlow, data preprocessing/epoching, and selecting evaluation metrics
- Will train an appropriate model in TensorFlow to categorize EEG data from a publicly available dataset

ADDITIONAL WORK EXPERIENCE

Prohibition Kitchen Port Jeff

Busser (Part-Time)

Port Jefferson, NY

Aug 2026 - Present

- Support all areas of front-of-house operations, consistently managing competing priorities in a fast-paced, team-driven environment

Circle M Beverage

Cashier (Part-Time) and Assistant Manager

Rocky Point, NY

Aug 2020 - Dec 2025

- Ran the store in the manager's absence, including opening/closing, running the store, and overseeing staff
- Trained customers to use rented equipment and troubleshoot equipment issues over the phone
- Drafted reference guides for other staff and customers regarding product selection and computer systems